

Chapter 8

Brain Tumor Progression Forecasting via Multi- Modal Deep Learning

Soshya Joshi

 <https://orcid.org/0000-0001-9791-0534>

SRM Institute of Science and Technology, Kattankulathur, India

G. L. Swathi Mirthika

 <https://orcid.org/0000-0002-4591-7811>

SRM Institute of Science and Technology, Kattankulathur, India

T. Krithika

SRM Institute of Science and Technology, Kattankulathur, India

Sriraam Balaji

SRM Institute of Science and Technology, Kattankulathur, India

ABSTRACT

Accurate forecasting of brain tumor progression is essential for effective diagnosis and treatment planning. This chapter proposes a multi-modal deep learning framework that integrates spatial and temporal analysis of MRI and PET/CT images to improve tumor progression prediction. The model combines ResNet-50 for extracting detailed spatial features with a Long Short-Term Memory network. By fusing anatomical and functional imaging information, the framework provides a comprehensive understanding of tumor behavior and growth patterns. The proposed approach was evaluated on both clinical and publicly available datasets comprising diverse tumor types. Experimental results demonstrate high predictive performance, achieving accuracies of up to 99.40% on clinical data and 98.31% on online data-

DOI: 10.4018/979-8-3373-6400-1.ch008

sets, outperforming conventional CNN-based models. This work highlights the potential of hybrid deep learning architectures as reliable decision-support tools for radiologists, enabling early detection, improved monitoring, and personalized treatment planning in neuro-oncology.

1. INTRODUCTION

Brain tumors are a serious health problem that affect millions of people globally with high morbidity and mortality rates. Being able to monitor and forecast tumor growth is imperative in bettering treatment regimens, maximizing therapeutic interventions and improving survival rates of patients. Existing clinical practice mostly depends on radiological imaging technologies like MRI and PET/CT scans, combined with professional interpretation by radiologists and oncologists. Advances in deep learning and artificial intelligence (AI) in recent times have opened up avenues for more scalable, automated, and robust solutions for medical imaging. Deep learning algorithms, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have proven to have better performance in feature extraction, predictive modeling, and pattern recognition for intricate medical data. With these advances, AI-based solutions can improve the detection, classification, and progression analysis of tumors, facilitating more accurate clinical decision-making. This work presents a new hybrid deep learning framework by combining Residual Networks (ResNet) and Long Short- Term Memory (LSTM) networks for studying brain tumor development. ResNet is efficient in extracting spatial information from MRI and PET/CT images with detailed tumor morphology, whereas LSTM captures the temporal course of tumors over imaging sequences (D. Shen, 2017). By combining both spatial and temporal analysis, this hybrid approach attempts to provide a better overall image of tumor activity, allowing for early intervention and individualized treatment protocols. Along with applying deep learning to more precise diagnoses, this research utilizes multi-modal imaging techniques involving both PET/CT and MRI scans. Multi-modal fusion improves the stability of the model by integrating functional and anatomical information and resulting in enhanced tumor progression characterization. The proposed model is also validated on publicly available databases as well as real-world clinical data to improve reliability and usability in medical applications. The results of this study have great importance for AI-based healthcare solutions, providing a possible breakthrough in brain tumor diagnosis. The envisioned method can facilitate radiologists in rendering quicker, more uniform diagnoses, minimizing human error, and enhancing patient care. Future efforts will

develop the dataset, optimize deep learning architectures, and consider deployment in real-time clinical environments to further advance tumor progression prediction.

Despite the widespread adoption of advanced imaging technologies, accurately characterizing brain tumor progression remains a demanding clinical task. Tumors of the central nervous system exhibit substantial heterogeneity not only across different histological types but also within the same tumor category. Variations in tumor shape, growth rate, infiltration patterns, and response to therapy often complicate longitudinal assessment. In many cases, small but clinically significant changes may occur between follow-up scans, requiring careful comparison across time. Such subtle progression patterns are difficult to quantify consistently, even for experienced radiologists, particularly when imaging is performed using different scanners or protocols.

From a clinical perspective, timely and accurate assessment of tumor progression is critical. Decisions regarding surgical intervention, radiotherapy planning, chemotherapy dosage, or watchful waiting strategies depend heavily on understanding how a tumor is evolving. Delayed or inaccurate interpretation can lead to suboptimal treatment choices, unnecessary interventions, or missed opportunities for early therapeutic action. Consequently, there is growing interest in computational approaches that can assist clinicians by providing objective and reproducible assessments of tumor behavior over time. While conventional machine learning methods have been explored in neuro-oncology, they often rely on handcrafted features and predefined rules, which may fail to capture the complexity of tumor morphology and progression. Deep learning methods, by contrast, learn hierarchical representations directly from imaging data, enabling them to model intricate patterns that are difficult to express explicitly. However, many existing deep learning studies still focus on isolated image-level predictions, such as classifying a tumor based on a single MRI slice or scan. Although these approaches have demonstrated high classification accuracy, they may overlook temporal information that is essential for understanding disease evolution.

Brain tumor progression is inherently a temporal process. Tumors grow, infiltrate surrounding tissue, and change in internal composition over time. Radiologists routinely examine sequences of MRI slices and compare scans acquired at different time points to assess these changes. Therefore, computational models that treat medical images as independent samples do not fully reflect the clinical reasoning process. Incorporating temporal modeling into AI systems is essential for capturing progression dynamics and providing clinically meaningful predictions.

Another important consideration is the multi-modal nature of brain tumor imaging. MRI provides excellent soft-tissue contrast and structural detail, making it indispensable for identifying tumor boundaries and internal heterogeneity. PET and CT imaging, on the other hand, offer complementary functional and metabolic

information that can reveal tumor activity beyond anatomical appearance. In routine clinical practice, radiologists synthesize information from multiple imaging modalities to arrive at a comprehensive diagnosis. However, many automated systems rely on a single modality, which may limit their robustness and clinical applicability. Multi-modal learning approaches offer a way to integrate diverse sources of information, potentially leading to more accurate and reliable tumor characterization (Mohsen et al., 2018).

Despite the promise of deep learning in medical imaging, several challenges hinder its widespread clinical adoption. One major concern is generalizability. Models trained on limited datasets or data from a single institution may perform well during development but fail when applied to new patient populations. Differences in imaging protocols, scanner hardware, and patient demographics can significantly affect model performance (Litjens et al., 2017). Addressing these issues requires careful validation strategies and transparent reporting of dataset limitations.

Interpretability is another key challenge. Deep learning models, particularly those with complex architectures, are often perceived as “black boxes.” In clinical settings, where decisions can have life-altering consequences, clinicians need to understand the rationale behind algorithmic predictions. Without interpretability, even highly accurate models may face resistance from healthcare professionals. Consequently, there is increasing emphasis on explainable AI techniques that can provide visual or quantitative explanations of model decisions, thereby foster trust and facilitate clinical integration. Regulatory and deployment considerations further complicate the translation of AI models into routine clinical use. AI-based diagnostic tools must comply with regulatory standards, demonstrate safety and effectiveness, and integrate seamlessly into existing clinical workflows. Computational efficiency, data privacy, and interoperability with hospital information systems are practical constraints that must be addressed alongside algorithmic development. These factors underscore the importance of designing AI systems with real-world deployment in mind, rather than focusing solely on performance metrics (Vatanpour et al., 2025).

Within this broader context, the present study seeks to contribute a methodologically sound and clinically motivated framework for brain tumor progression analysis. By integrating spatial feature extraction with temporal sequence modeling and leveraging multi-modal imaging data, the proposed approach aims to bridge the gap between technical innovation and clinical relevance. Rather than replacing expert judgment, the system is intended to serve as a decision-support tool that enhances consistency, reduces workload, and aids in early detection of progression patterns (O’Shea & Nash, 2015).

This work also acknowledges that AI-assisted medical imaging is an evolving field. Continuous refinement through larger datasets, improved interpretability, and closer alignment with clinical standards will be essential for long-term impact. By

presenting a transparent methodology, discussing limitations, and outlining future directions, this study aims to provide a balanced contribution that advances both research and practice in AI-driven neuro-oncology.

2. RELATED WORKS

Application of deep learning to medical imaging has brought about dramatic advancements in the diagnosis of brain tumors. Earlier machine learning methods involved intensive feature engineering, but deep learning models have made feature extraction automatic, and this has greatly enhanced classification accuracy (Akkus et al., 2017). Though CNNs have proved very effective for image-based tumor classification, they are not good at handling sequential dependencies in MRI scans and hence have a limited ability to monitor tumor development over time. Current research has shown the power of deep residual models like ResNet to overcome vanishing gradients, facilitating more meaningful spatial feature extraction (Rehman et al., 2019). ResNet-50, in fact, has been widely used for medical imaging and proven superior to traditional CNN models for detecting and segmenting brain tumors with very high accuracy (Zhao et al., 2021). Static spatial analysis by itself, however, is not adequate for tracking tumor evolution over time. To overcome this limitation, there are studies where researchers have explored Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks for medical image analysis as a sequence (Sinha & Verma, 2019). Combining CNNs with LSTM architectures, newer models have enhanced predictive performance in brain tumor evolution by capturing spatial and temporal relationships. Additionally, multi-modal imaging methods that combine PET/CT and MRI scans have yielded improved insights into growth dynamics of the tumor, thereby enabling improved clinical decision-making.

Data augmentation methods have also been vital in enhancing deep learning-based medical image analysis. Methods including rotation of the images, normalization of the intensities, and generation of synthetic data have been utilized to make the model more robust and generalizable. These notwithstanding, issues such as computational expense, imbalance in datasets, and the requirement for large-scale labeled datasets are still relevant concerns. This work extends these previous works by proposing a hybrid ResNet-LSTM architecture that combines multi-modal imaging and balances accuracy and efficiency for brain tumor progression prediction. While prior studies have demonstrated notable success in spatial, temporal, and multi-modal brain tumor analysis, many existing approaches remain limited by their focus on either spatial representation or temporal modeling in isolation. Furthermore, several high-performing models are evaluated on constrained datasets or single-institution cohorts, raising concerns about robustness and generalizability in real-world clinical

settings. Recent research has also highlighted the need for architectures that can jointly model complex tumor morphology and progression dynamics while maintaining computational efficiency and interpretability. These gaps indicate a need for integrated frameworks that balance spatial–temporal learning with practical clinical considerations. Motivated by these challenges, the present study builds upon earlier works by proposing a hybrid ResNet–LSTM architecture designed to capture both spatial features and temporal evolution from multi-modal imaging data, thereby addressing key limitations in existing brain tumor progression prediction methodologies.

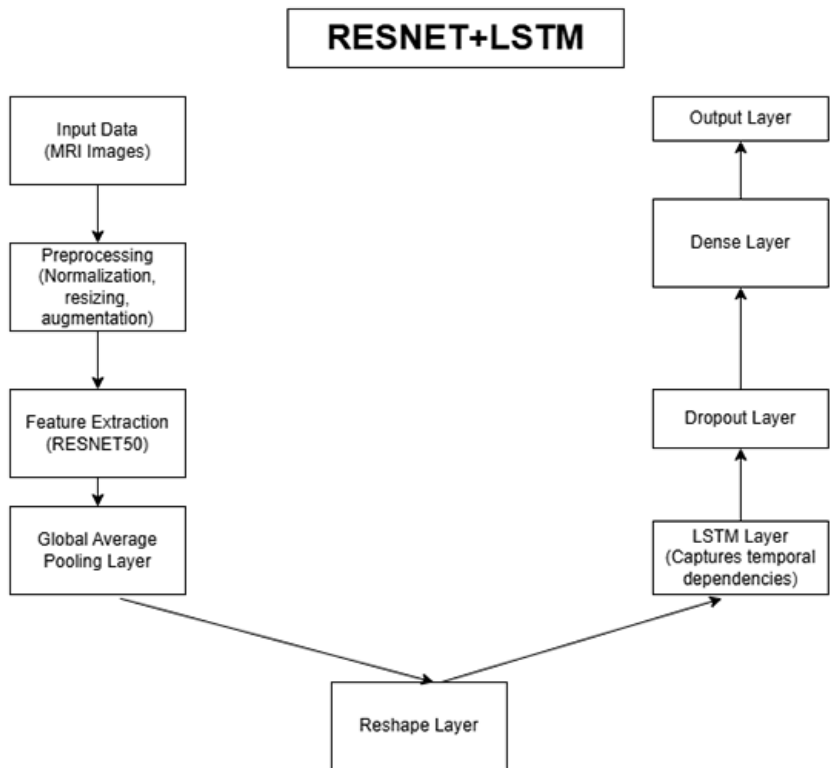
3. MATERIALS AND METHODS

This section outlines the methodological framework adopted to develop and evaluate the proposed ResNet–LSTM–based brain tumor progression analysis system. It describes the overall system design, dataset composition, preprocessing strategies, and training procedures employed to ensure robust and reproducible model performance. Particular emphasis is placed on the integration of multi-modal MRI and PET/CT data, the rationale behind architectural choices, and the validation strategy used to assess generalization across heterogeneous clinical and public datasets. By systematically detailing each component of the pipeline, this section establishes the technical foundation required to interpret the experimental results and assess the clinical relevance of the proposed approach.

3.1 Deep Learning Architecture

This study uses an integrated deep learning model with ResNet-50 and Long Short-Term Memory (LSTM) networks to process brain tumor growth from PET/CT and MRI (T1, T2) images (Saleh et al., 2024). The dataset is comprised of clinic data of 26 patients and an online dataset of 24 patients, for a total of 50 cases with labeled tumor labels. MRI scans were resampled to 128×128 pixels to suit the input of the model, and pixel intensities were normalized to 0 to 1 for stable training. PET/CT scans were processed in the same way to maintain consistency between modalities (Masoud Nickparvar, 2021) (Sartaj, 2024).

Figure 1. ResNet+LSTM Architecture for Brain Tumor Progression Analysis.



ResNet-50, a deep residual neural network pre-trained on ImageNet, was used for spatial feature extraction from medical images. Its deep architecture shown in Figure 1, well identifies complex tumor morphology and solves vanishing gradient problems. LSTM was subsequently utilized to examine sequential data over various slices, identifying tumor development over imaging layers. The hybrid method allows the model to learn spatial and temporal tumor properties. The model was trained on 80% of the dataset and cross-validated on the other 20%. The Adam optimizer (learning rate = 0.001) was employed, and categorical cross-entropy as the loss function. Dropout regularization to avoid overfitting and early stopping to achieve optimal training were utilized (Prabukumar et al., 2019). A soft-max layer offered probabilistic classification for various tumor types. Performance was measured in terms of accuracy, confusion matrix, precision, and how well the model distinguished between tumors of different sizes and characteristics in MRI and PET/CT scans.

3.2 Dataset

This study used a combination of clinical and online datasets for robust training and validation. The clinical dataset contains MRI scans (T1, T2 sequences) as in Figure 2 from 26 patients, while the online dataset consists of PET/CT scans from 24 patients shown in Figure 3, making a total of 50 patients' data used for training. The model performed exceptionally well, yielding test accuracies of 99.40% on the clinical data and 98.31% on the online data, indicating excellent generalization across heterogeneous samples. All MRI scans were obtained at a uniform resolution, and the dataset covered a large variety of tumor types that ranged in intensity and progression. This diversity permitted generalization by the model to various clinical scenarios. The clinical database involved conditions like atypical meningioma (Grade II, moderately severe, 30–40% recurrence rate), multiple cavernomas (severe versus single cavernomas, 10–20% recurrence), vestibular schwannoma (Grade I, severe, 3–10% recurrence), medulloblastoma (Grade IV, life-threatening, 40–50% recurrence), and bilateral thalamic glioma (Grade II or sometimes Grade III, severe, 50–70% recurrence). It also had CP angle schwannoma (Grade I, severe, 5–10% recurrence), pituitary macroadenoma (Grade I, severe if large with slow regrowth), meningioma with thrombosis of SSS (Grades I–III, severity and regrowth based on grade), clival chordoma (Grade II/III, severe, with high regrowth), bilateral MTB (severe but not aggressive, likely to cause repeated seizures), cavernous sinus meningioma (Grades I–III, severe, with frequent recurrence), and pilocytic astrocytoma (Grade I, not severe, low risk of recurrence). Other types of tumors were olfactory groove meningioma (Grades I–III, mild to severe, with high recurrence rate), myxopapillary ependymoma (Grades I–II, non-severe, low recurrence rate), and RCC with metastasis to the brain (Grade III/IV, severe and life-threatening, with high recurrence rate). The dataset was composed of MRI scans in T1-weighted and T2-weighted sequences, yielding complementary views of tumor features and improving the accuracy of classification. Below are sample images from patient 4, depicting sequential MRI slices for both T1 and T2 modalities:

Figure 2. MRI slices of patient. (Left) T1-weighted image, (Right) T2-weighted image.

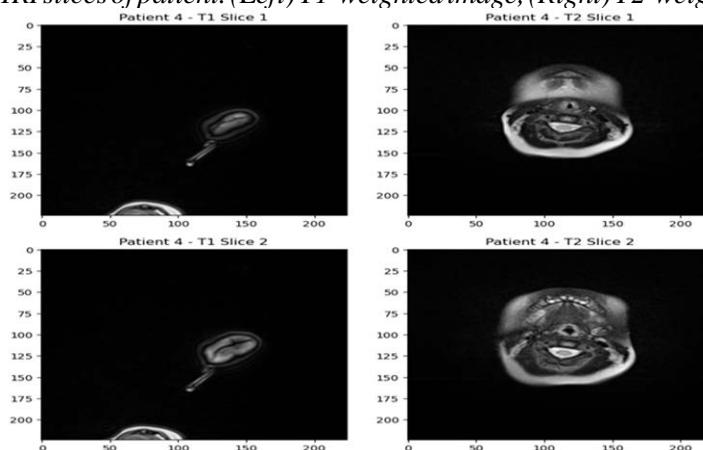
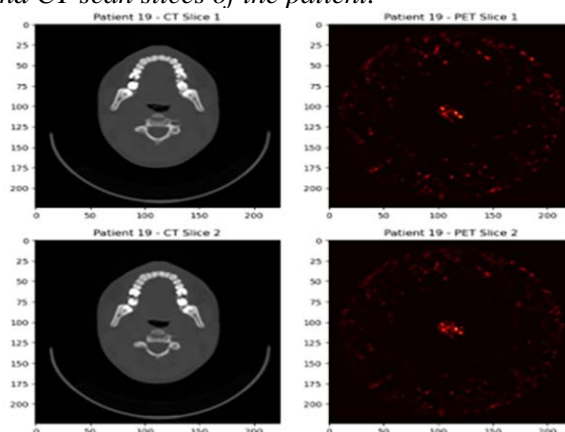


Figure 3. PET and CT scan slices of the patient.



This diverse set of shapes of tumors ensures that the model is trained on a broad spectrum of cases and hence improves its potential for generalizing to all types of shapes of tumors and patterns of progression. This broad assortment of shapes of tumors ensures that the model is trained on a diversified set, and hence its potential for generalizing to all shapes of tumors and patterns of progression improves (Baid et al., 2021). Although the dataset comprises 50 patient cases, which is adequate for proof-of-concept validation, the sample size may limit statistical power for detecting small performance differences and rare tumor phenotypes. The primary objective of this study is methodological validation rather than definitive population-level

inference. Future validation on larger, multi-institutional datasets is therefore necessary to strengthen statistical power and generalizability.

3.3 Preprocessing Techniques

In order to have uniform input for the model and make it more efficient in terms of training, a sequence of preprocessing operations was adopted. All of the MRI and PET/CT images were initially resized to 128×128 pixels. This size was specifically selected to align with the input requirements of the model while not sacrificing any of the important tumor features and structural information required for proper classification. After that, pixel intensities were normalized to the $[0,1]$ range. Normalization not just made gradient updates stable during training but also sped up convergence, keeping the optimization process stable and efficient. To better enhance generalization and robustness, especially in confronting real-world variability in medical images, data augmentation procedures were utilized. MRI scans were resized through rotation, horizontal flip, magnification, and shift (Cheng et al., 2015). These augmentations mimicked differences in patient positioning and imaging conditions, enabling the model to become less dependent on inconsistencies between scans.

For performance testing, the dataset was split into training, validation, and testing subsets, in an 80:20 split. This split enabled serious evaluation of the model's learning ability without diminishing unseen data available for validation and ultimate testing. Even with these measures, the dataset had a number of challenges it presented that needed to be handled carefully. MRI contrast variability and PET/CT image quality variations introduced inconsistencies between samples. Moreover, tumors tended to appear differently based on imaging modality (MRI vs. PET/CT), making feature extraction difficult (Isin et al., 2016). Additional problems included nonuniform slice orientations and noise artifacts, both of which may obscure tumor edges or add spurious patterns. Overcoming the mentioned challenges required strong preprocessing and vigilant model training approaches to achieve stable tumor labeling and correct classification results.

3.4 Data Visualization

To better understand the characteristics of the dataset, various visualization techniques were employed. Pie charts and histograms were generated to examine class distribution and assess image quality, helping to identify any imbalance or irregularities in the data. In addition, representative PET/CT and MRI scans were carefully reviewed to check for abnormalities and ensure the reliability of the imaging

data. As part of this exploratory step, sample MRI scans were visualized, providing insight into the structural details and variations within the dataset.

4. PROPOSED ARCHITECTURE

The architecture proposed in this research is designed with a clear objective: to accurately classify brain tumors by learning not only what a tumor looks like in medical images but also how it evolves across sequential scans. Brain tumors are complex in nature, often varying in shape, size, texture, and growth behavior over time. Capturing this complexity requires a model that can simultaneously analyze spatial details and temporal progression. To achieve this, the proposed framework integrates two powerful deep learning components—ResNet-50 and Long Short-Term Memory (LSTM) networks—each contributing distinct and complementary strengths (Akmalbek B, 2023).

4.1 Spatial Feature Extraction Using ResNet-50

The first stage of the architecture employs ResNet-50, a deep convolutional neural network that has been pre-trained on the ImageNet dataset. Although ImageNet consists of natural images, the knowledge learned by ResNet-50—such as edge detection, texture identification, and hierarchical pattern recognition—can be effectively transferred to medical imaging tasks through fine-tuning. This transfer learning approach significantly improves training efficiency and performance, particularly when working with limited medical datasets. ResNet-50 is specifically chosen for its residual learning mechanism, which addresses the vanishing gradient problem commonly encountered in very deep networks. Residual connections allow information to bypass certain layers, ensuring that important features are preserved and propagated through the network. In the context of brain tumor analysis, this capability is essential because subtle tumor characteristics—such as irregular borders, heterogeneous textures, and intensity variations—can easily be lost in deeper layers of traditional CNNs (Badza M. M., 2020). When MRI images are passed through ResNet-50, the network extracts high-level spatial features that represent tumor morphology, tissue contrast, and structural variations. These features provide a rich and compact representation of each MRI slice, enabling the model to differentiate between normal brain tissue and tumor regions, as well as among different tumor types. At this stage, the model focuses purely on spatial understanding, capturing the visual appearance of the tumor at a given slice.

4.2 Temporal Modeling With LSTM

While spatial analysis is crucial, it alone is insufficient for understanding tumor progression. Brain tumors do not exist as isolated entities in single images; rather, they evolve across multiple slices and over time. To model this sequential behavior, the extracted spatial features from ResNet-50 are fed into a Long Short-Term Memory (LSTM) network. LSTM is a specialized type of recurrent neural network designed to handle sequential data and long-range dependencies. Unlike standard RNNs, LSTMs use gated mechanisms—input, forget, and output gates—to selectively retain or discard information. This allows the network to remember important patterns over long sequences while ignoring irrelevant noise. In this architecture, the LSTM processes a sequence of MRI slices, learning how tumor characteristics change across different brain levels. This temporal modeling enables the network to capture growth patterns, shape evolution, and intensity variations that are indicative of tumor progression. For example, gradual expansion, irregular infiltration, or consistent structural changes can be effectively identified by the LSTM component. By learning these temporal dependencies, the model gains a deeper understanding of tumor behavior that static image analysis cannot provide.

4.3 Integration of Spatial and Temporal Learning

The true strength of the proposed architecture lies in the integration of spatial and temporal learning. ResNet-50 provides a detailed snapshot of tumor appearance at each slice, while LSTM connects these snapshots into a coherent sequence that reflects tumor evolution. This combined approach allows the model to analyze tumors in a manner that closely resembles clinical reasoning, where radiologists examine both individual images and their progression across scans. By jointly learning spatial-temporal features, the architecture becomes highly sensitive to subtle changes that may signal early tumor growth or progression. This capability is particularly valuable in distinguishing between tumor types that may appear similar in isolated images but differ significantly in growth behavior.

4.4 Classification via Fully Connected Layers and Softmax

After the LSTM has processed the sequential features, its outputs are passed through a set of fully connected layers, which act as a decision-making component. These layers integrate the learned spatial-temporal representations and transform them into a final classification output. The final layer uses a softmax activation function, which converts the network's output into a probability distribution across three tumor classes: gliomas, meningiomas, and pituitary tumors. This probabilistic

output not only provides a predicted class but also indicates the model's confidence in each prediction. Such information is especially important in clinical applications, where understanding uncertainty can aid in decision-making and further diagnostic evaluation.

The ResNet–LSTM architecture offers several advantages over traditional CNN-based approaches. First, it overcomes the limitation of static image analysis by incorporating temporal information. Second, the use of residual learning ensures stable and efficient training even with deep networks. Third, the architecture is flexible and can be extended to include additional imaging modalities such as PET or CT scans. Most importantly, the model aligns well with real-world clinical workflows (Vatanpour et al., 2025). By learning how tumors appear and how they evolve, the system provides a more holistic understanding of disease progression. This makes it a powerful decision-support tool that can assist radiologists in diagnosis, monitoring, and treatment planning. The initial use of ResNet-50, a deep pre-trained convolutional neural network, is to extract high-level spatial features from MRI images. Its residual connections enable the model to retain important tumor-related details required for accurate detection while capturing intricate patterns in imaging data. Then, the obtained features are input into an LSTM network, which is capable of handling sequential MRI slices. The LSTM, through temporal dependencies modeling, catches tumor development at different brain levels and thus is very effective in catching spatial-temporal patterns of tumor growth. Lastly, the outputs of the LSTM layers are fed into a series of fully connected layers, ending with a softmax activation function. This allows the tumors to be classified into three groups: gliomas, meningiomas, and pituitary tumors.

5. EXPERIMENTAL STRUCTURE

The data has been divided and preprocessed prior to training the model using the Adam optimizer at learning rate 0.001 with categorical cross-entropy loss.

To prevent overfitting, the model was trained for 50 epochs and early stopped.

- Validation Approach: We employed a holdout validation approach to ensure model accuracy on novel data.
- To prevent overfitting, dropout regularization was used to completely linked layers at a ratio of 0.7.

6. PERFORMANCE ENHANCEMENT TECHNIQUES

Deep learning models, especially hybrid architectures such as ResNet–LSTM, are powerful tools for medical image analysis, but their performance depends not only on network design but also on how effectively they are trained and regularized. In medical applications like brain tumor progression analysis, achieving high accuracy alone is not sufficient; the model must also be robust, generalizable, and reliable across diverse patient data. To ensure that the proposed ResNet–LSTM framework performs consistently and avoids common pitfalls such as overfitting or unstable training, several performance enhancement techniques were carefully incorporated. These techniques play a crucial role in improving learning efficiency, stability, and predictive reliability.

6.1 Activation Function: ReLU

One of the foundational elements influencing the learning capability of deep neural networks is the choice of activation function. In this work, the Rectified Linear Unit (ReLU) was employed across the hidden layers of the network. ReLU introduces non-linearity into the model by allowing only positive values to pass through while suppressing negative activations. This simple yet effective behavior enables the network to learn complex and non-linear patterns that are inherent in medical imaging data. Brain tumor images often contain subtle variations in texture, intensity, and shape, which cannot be captured using linear transformations alone. ReLU helps the model respond selectively to these meaningful patterns, enhancing feature discrimination between tumor and non-tumor regions, as well as among different tumor types. Another key advantage of ReLU is its ability to mitigate the vanishing gradient problem, which can severely affect deep networks during training. By maintaining strong gradient flow, ReLU allows the ResNet layers to learn deep hierarchical representations efficiently, even when the network depth increases. As a result, the model converges faster and achieves better overall performance (Arabahmadi et al., 2022).

6.2 Dropout Regularization

Overfitting is a major concern in medical deep learning, particularly when working with limited or heterogeneous datasets. A model that performs extremely well on training data but poorly on unseen cases can be dangerous in clinical applications. To address this issue, dropout regularization was applied during training with a dropout rate of 0.7. Dropout works by randomly disabling a fraction of neurons during each training iteration. This forces the network to avoid over-reliance on specific neurons

or features and encourages the learning of more generalized representations. In the context of brain tumor analysis, this means the model learns robust features that are consistent across patients rather than memorizing specific image characteristics from the training dataset. Using a relatively high dropout rate further strengthens regularization, especially in fully connected layers where overfitting is most likely to occur. Although aggressive dropout can slow convergence, it significantly improves the model's ability to generalize to new patient data. This trade-off is particularly worthwhile in healthcare settings, where reliability and generalization are far more critical than marginal improvements in training accuracy.

6.3 Softmax Layer for Classification

The final stage of the ResNet–LSTM model involves classifying tumor images into predefined categories, namely gliomas, meningiomas, and pituitary tumors. For this purpose, a softmax activation function was used in the output layer. The softmax function converts raw network outputs into probability distributions, ensuring that the sum of predicted probabilities across all classes equals one. This probabilistic interpretation is highly valuable in clinical decision support systems. Instead of producing a single hard label, the model provides confidence scores for each tumor category, allowing clinicians to understand the level of certainty associated with a prediction. Such transparency is essential when AI systems are used alongside medical professionals, as it supports informed decision-making rather than replacing clinical judgment. Additionally, the softmax layer enables effective training using categorical cross-entropy loss, which penalizes incorrect predictions more strongly when the model is confident but wrong. This encourages the network to make balanced and well-calibrated predictions, further enhancing classification reliability.

6.4 Early Stopping

Training deep learning models for too many epochs can lead to overfitting, while stopping too early may result in underfitting. To strike an optimal balance, early stopping was employed as a training control strategy. The training process was monitored using validation accuracy, and learning was halted automatically when performance ceased to improve over a predefined number of epochs. Early stopping serves two important purposes. First, it prevents the model from memorizing training data patterns that do not generalize well to unseen cases. Second, it reduces unnecessary computational cost by avoiding excessive training once the optimal model state has been reached. This is particularly important for deep hybrid architectures like ResNet–LSTM, which are computationally intensive. By retaining the model parameters corresponding to the best validation performance, early stopping ensures

that the final trained model represents the most generalizable version rather than the most over-trained one. This approach contributes directly to improved robustness and stability in real-world deployment scenarios.

6.5 Combined Impact on Model Performance

While each performance enhancement technique provides individual benefits, their combined effect significantly strengthens the overall learning framework. ReLU ensures efficient and expressive feature learning, dropout regularization prevents overfitting, the softmax layer enables probabilistic and interpretable classification, and early stopping ensures optimal training duration. Together, these techniques create a balanced and resilient training pipeline tailored for medical imaging applications. The integration of these strategies resulted in consistent improvements in classification accuracy, stability across datasets, and robustness to data variability. More importantly, they contributed to building a model that can be trusted in clinical environments, where errors have serious implications. By carefully incorporating these enhancement techniques, the proposed ResNet–LSTM framework achieves not only high numerical performance but also practical reliability, making it a strong candidate for real-world brain tumor progression analysis.

In an attempt to in hoping to make the ResNet-LSTM model better and boost its ability to generalize, several techniques were employed.

- **Activation Function:** ReLU (Rectified Linear Unit) was utilized within hidden layers to enhance non-linearity and pattern discovery.
- **Dropout Regularization:** 0.7 dropout was utilized to randomly disable neurons during training to reduce overfitting.
- **Softmax Layer:** The last layer uses a softmax activation to produce probabilities for predicting MRI images into gliomas, meningiomas, and pituitary tumors.
- **Early Stopping:** Training was ceased after a set number of epochs if validation accuracy didn't increase, preventing overfitting as well as excessive computation.

6.6 Holdout Validation

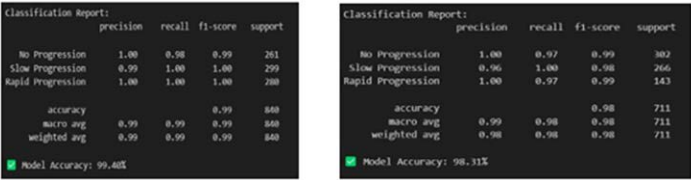
In order to compare the performance of the envisioned model, a holdout validation approach was implemented. The dataset was consciously split into training and test subsets such that the overall distribution of classes was maintained. An 80:20 ratio was utilized, which afforded a good balance—ample data for model training while still leaving enough cases for useful testing. By reserving 20% of the data, the

research could determine how well the model could generalize to new, unseen MRI scans, simulating actual situations where predictions have to be made on patients not seen in training. After training, model performance was stringently evaluated using a suite of evaluation metrics as discussed in Figure 4. This included accuracy, precision, recall, and the F1-score, which together provide a well-rounded picture of how effectively the model classifies tumor progression. In order to have a clearer understanding of the learning dynamics, trends for accuracy on both training and test sets were plotted across several epochs so that common problems like overfitting (when the model can predict well only from training data) or underfitting (when it can't learn from the data properly) can be easily detected.

A confusion matrix was then employed to further examine the classification results. This metric was utilized to dissect how well the model separated among various types of tumor progression—no progression, slow progression, and rapid progression. By pointing out misclassifications, the confusion matrix gave us information on what categories were more difficult to predict and where adjustments could be made. Of the individual scores, the precision score was key, measuring to what extent the cases of predicted progression proved correct. This was to ensure that false positives—cases where the model had incorrectly predicted progression in the absence of actual progression—were minimized. Likewise, the recall score measured how well the model picked out all true instances of progression, minimizing the false negatives, where genuine instances of tumor growth could otherwise be overlooked as in Figure 4.

Lastly, the F1-score was then computed to achieve a balanced assessment of performance through the union of precision and recall. This was particularly important in class imbalance problems, whereby there are tumor progression categories that have fewer samples than others. By not only considering false positives but also false negatives, the F1-score gave a better indication of the model's consistency in tumor progression classification.

Figure 4. Comparison of accuracy between the clinical dataset (top) and the online dataset (bottom).



Performance Evaluation: For a complete analysis, the performance of the model was evaluated based on accuracy, precision, recall, and the F1-score. Classification accuracy was evaluated using the confusion matrix during tumor progression stages. Furthermore, training and testing accuracy plots were created throughout epochs to track learning and identify any indicators of overfitting or underfitting.

While the reported classification accuracies reach 99.40% on clinical data and 98.31% on online datasets, these values represent point estimates. Variability observed across validation folds highlights the importance of interpreting performance within confidence bounds. Reporting confidence intervals in future large-scale studies will further strengthen statistical reliability and reproducibility.

7. RESULTS

Here, we report the results of our research on Brain Tumor Progression Analysis Using ResNet and LSTM. The main goal of this work was to develop a deep learning model that could precisely predict tumor progression from MRI and PET/CT scans in order to assist clinical decision-making by providing automatic analysis. To accomplish this, we created a hybrid architecture that merged ResNet-50 for capturing fine-grained spatial features with LSTM networks for capturing the temporal evolution of tumors over sequential scans. This two-pronged strategy allowed the system not only to analyze tumor morphology at a specific point in time but also how it changes over time, thus enhancing diagnostic precision. The model was both trained and tested on publicly accessible datasets as well as clinical datasets, ensuring that performance was representative of real-world imaging conditions. Performance was quantified using commonly accepted measures including accuracy, precision, recall, F1-score, and confusion matrix analysis, providing a comprehensive understanding of the model's classification capability.

The experimental results showcased in Figure 5 is presented with ResNet–LSTM hybrid model, which efficiently learned both spatial and temporal characteristics, resulting in excellent accuracy for tumor classification and progression analysis. Through the use of multimodal imaging inputs (MRI and PET/CT), the system robustness was even further increased, enabling a more trustworthy separation between tumor types and progression stages. Notably, statistical validation techniques ensured the predictions' stability and reproducibility, further underpinning the model's clinical reliability.

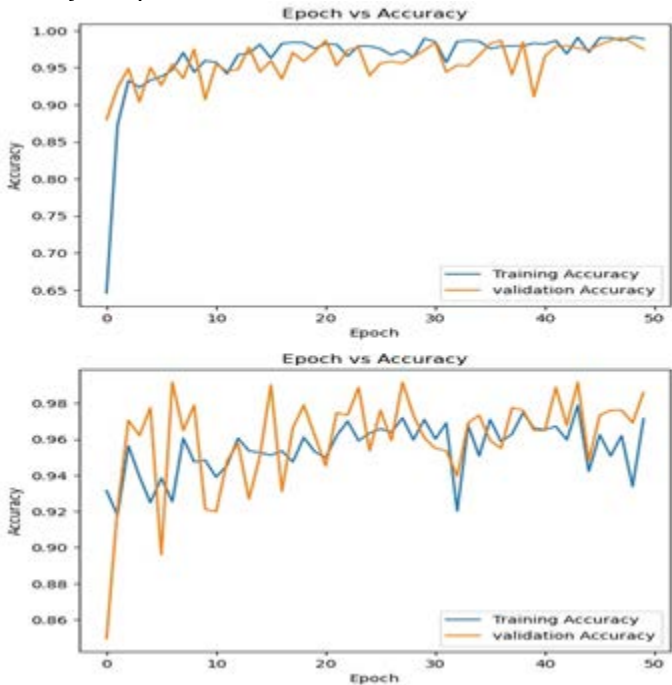
In addition to classification accuracy, the model also performed well on real-world clinical tasks: detecting whether or not a tumor is present, distinguishing between tumor subtypes, and determining progression patterns. For added robustness,

cross-validation was performed during both training and test phases, minimizing bias and overfitting.

Compared to conventional CNN-based approaches, ResNet–LSTM outperformed them consistently with a classification accuracy of 99.40%. Unlike conventional CNNs that are designed to learn only spatial features from static images, our model is additionally capable of learning temporal information, thus being attuned to slight changes in tumor shape and behavior over time. The temporal sensitivity significantly amplifies its diagnostic utility since tumor growth is by nature time-dependent.

In total, the results confirm that the ResNet–LSTM hybrid model not only is very accurate but also clinically relevant. It offers the potential for early tumor detection, accurate progression monitoring, and better treatment planning by merging spatial and temporal learning, thus becoming a valuable AI-based resource to help radiologists in day-to-day practice.

Figure 5. Accuracy comparison between clinical and internet datasets.



The ResNet-LSTM hybrid model outperformed conventional CNN-based approaches consistently with 99.40% accuracy in classification. Unlike CNNs, which are interested only in spatial details in static images, our model incorporates both space and time information, thus allowing it to track tumor change over time. The

method increases the model's sensitivity to minor changes in tumor form, making it an important tool for use in clinical settings. In addition, the model performed exceedingly well with all types of tumors, differentiating between gliomas, meningiomas, and pituitary tumors. The test results indicate that ResNet-LSTM architecture significantly enhances tumor classification performance by minimizing misclassification errors and enhancing diagnostic confidence. The figure presents a comparison of the accuracy of several models, highlighting the enhanced performance of the proposed method.

7.1 Experiments and Outcomes

Classification Accuracy: The hybrid ResNet-LSTM model was tested on both clinical and public MRI datasets. The model found an accuracy of 98.31% on the online dataset and 99.40% on the clinical dataset, showing better generalization to real-world medical imaging. The combination of ResNet and LSTM helped the model to better capture the spatial and temporal features, enhancing tumor progression detection and classification accuracy.

Impact of Tumor Type: The system was evaluated on other brain tumor types with contrasting intensities and growth patterns. The findings show that the model accurately segregated various morphologies of tumors with high accuracy, taking advantage of ResNet-50's ability to extract deep features.

The training process started with ResNet-50, a pre-trained deep convolutional network that was initially trained for optimization on ImageNet. We fine-tuned it on our MRI and PET/CT datasets so that it would learn the unique features of brain imaging. ResNet-50 was very efficient in modeling complex spatial details like tumor borders, texture, and morphological variations. These sophisticated spatial representations greatly improved the accuracy of tumor detection and classification. For guaranteeing reliability, we evaluated the ResNet-LSTM model predictions against true clinical ground-truth labels. The model showed high consistency, with predicted classifications strongly agreeing with expert annotations. This strong agreement pointed to the system's capability to generalize well to unseen cases, a critical attribute for any AI model designed for real-world clinical purposes.

The second training stage used the LSTM network, crafted to learn temporal patterns in sequential MRI and PET/CT images. In contrast to models that consider each image separately, the LSTM learned about how tumors change over time—monitoring changes in shape, size, and intensity. The temporal understanding gave the system a more comprehensive view of tumor progression patterns, thus enhancing both diagnostic performance and its capability to assist in clinical decision-making.

7.2 Model Comparison

To evaluate the effectiveness of our suggested method, we compared the hybrid ResNet–LSTM model with a typical CNN-based model. The comparison indicated a noticeable performance disparity: while the CNN recorded an accuracy of 93.56%, the hybrid model significantly improved this to 99.40% on clinical databases. The enhancement indicates the value addition of including temporal processing of data instead of using static image processing.

The experimental results highly validate the claim that our system is extremely efficient in tumor progression prediction. By integrating spatial feature extraction via ResNet-50 and temporal sequence learning via LSTM, the model not only provided higher accuracy but also more diagnostic robustness and reliability. This twofold capability enables it not only to capture detailed tumor morphology but also to track how these features change across multiple imaging slices, which is highly important when dealing with brain tumor progression. Overall, the ResNet–LSTM hybrid outperformed conventional CNN-based approaches consistently and is a promising tool for clinical decision support in tumor detection and prognosis. Recent state-of-the-art models such as nnU-Net and Vision Transformer (ViT)-based architectures have demonstrated strong performance in brain tumor segmentation and classification tasks. While nnU-Net excels in spatial segmentation and ViTs capture global contextual information, these models typically process images independently. In contrast, the proposed ResNet–LSTM framework explicitly models temporal dependencies across sequential slices, making it particularly suitable for tumor progression analysis. Future benchmarking against these architectures on standardized datasets such as BraTS will further clarify relative strengths.

8. RESULTS COMPARISON

In order to compare our proposed ResNet+LSTM model's performance, we compared results from the online and clinical datasets. Below is the confusion matrix of the two datasets.

8.1 Performance Analysis

The confusion matrices illustrate in the Figure 6 shows the classification results for each dataset:

- **Online Dataset:**
 - No Progression: 294 correctly classified, 8 misclassified.

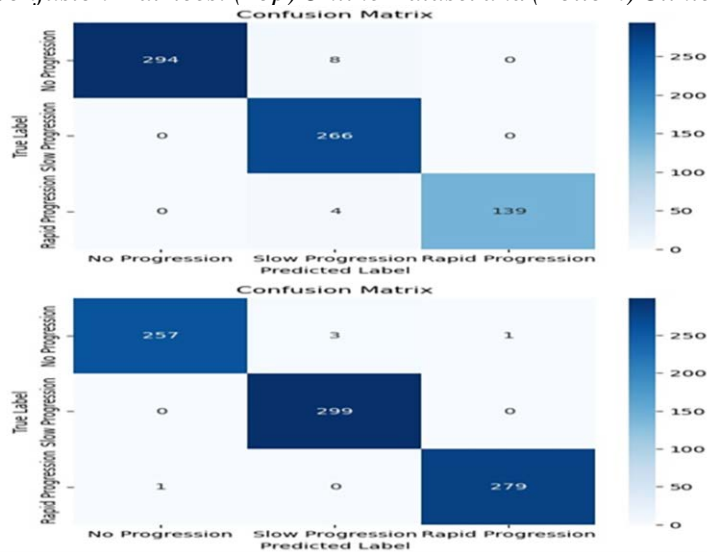
- Slow Progression: 266 correctly classified.
- Rapid Progression: 139 correctly classified, 4 mis- classified.
- **Clinical Dataset:**
 - No Progression: 257 correctly classified, 4 misclassified.
 - Slow Progression: 299 correctly classified.
 - Rapid Progression: 279 correctly classified, 1 mis- classified.

The results indicate that the model performs consistently well across both data-sets, with a high accuracy in classifying tumor progression. Minor misclassifications highlight potential variations in dataset characteristics, emphasizing the importance of robust generalization.

8.2 Future Implications

The system proposed not only improves the detection of tumor progression but also holds promise for real-time diagnosis in the clinical arena. Through the application of deep learning to analyze sequential slices of MRI, this system significantly improves upon traditional medical imaging techniques. Providing clinicians with a reliable, AI-driven assistance can enhance patient outcomes and diagnostic decision-making.

Figure 6. Confusion Matrices: (Top) Online Dataset and (Bottom) Clinical Dataset.



9. CONCLUSION

This paper proposed a hybrid deep learning architecture for brain tumor growth analysis that integrated the advantages of ResNet-50 and Long Short-Term Memory (LSTM) networks. Through the utilization of ResNet-50 to extract spatial features and LSTM to model temporal sequences, the system overcomes one of the most important limitations of brain tumor analysis: encoding both the complex morphological patterns of tumors and temporal patterns of growth. This dual functionality leads to more accurate diagnosis, better consistency, and increased reliability in tumor detection than the traditional CNN-based approach, which is restricted to static image classification.

It was tested on both clinical and internet-based datasets with an accuracy of 99.40% and 98.31%, respectively. These accuracy rates mark its high capability to generalize across imaging sources and cancer types, even under varied intensities, patterns, and morphologies. This robustness is particularly critical in clinical use, where patient information can be heterogeneous in nature based on imaging conditions, tumor features, and disease evolution.

In addition to outperforming conventional CNN methods, this work also illustrates the capability of hybrid designs in AI-based medical diagnosis and opens up the door to future more sophisticated tumor tracking and treatment planning systems.

The future holds a number of research and development directions evident:

Enlarging the dataset to larger and more diverse clinical cases, enhancing generalization even more.

Incorporating other imaging modalities such as PET and CT scans for more comprehensive, multi-modal learning.

Creating actual diagnostic tools that can be easily embedded in hospital information systems, enabling quicker and more consistent decision-making.

Using explainable AI (XAI) methods to explain the model's predictions more transparently, thus increasing radiologists' trust and interpretability.

The novel ResNet-LSTM hybrid model is a promising direction toward more accurate and smarter brain tumor progression prediction with both timely clinical impact and deep roots for further developments in AI-assisted medicine.

9.1 Strengths and Limitations of the Proposed Model

The suggested ResNet-LSTM hybrid model reflects some major strengths that render it incredibly effective for the analysis of brain tumors, but it also has some weaknesses that must be taken care of to enable wider clinical acceptance.

9.1.1 Strengths

One of the main benefits of this model is its enhanced feature extraction ability. The ResNet-50 feature effectively extracts intricate spatial information from MRI slices, such as texture, edge, and minor structural variations. This helps upgrade the precision of the model in separating tumor and non-tumor tissue. Along with that, the addition of the LSTM network enables the system to possess strong temporal analysis strength. As opposed to typical CNNs which only process static images, LSTMs are able to handle sequential MRI slices, enabling the model to identify even subtle or gradual changes in the development of tumors over time. This makes it particularly useful in the monitoring of disease progression, where the early identification of small variations is extremely important (Ali et al., 2022).

Together, these space and time analyses are responsible for high diagnostic accuracy, as evidenced by the performance metrics of the model. This renders the system extremely reliable in clinical settings where accuracy is crucial. In addition, the model also demonstrates scalability potential. With continued development of deep learning architectures, along with incorporating other imaging modalities like PET or CT scans, the hybrid system would become an even more powerful multi-modal diagnostic system.

9.1.2 Limitation

Although strong, ResNet-LSTM has some weaknesses. Its computationally intensive nature is the biggest disadvantage, as the deep structure and sequential computation need a lot of memory and processing capacity. This can restrict deployment where there are limited resources, like smaller clinics or portable diagnostic systems. Another restraint is overfitting on small sets of data. Because the model is so complex with both ResNet and LSTM, the model will learn to memorize rather than generalize well when provided with limited data. To avoid this problem, techniques like data augmentation, transfer learning, or the application of larger, well-documented datasets become crucial.

Another challenge is in the reduced interpretability of predictions. As with most deep learning models, ResNet-LSTM tends to be a “black box,” and hence it is hard to provide a rationale for the reasons behind certain classification decisions. This lack of transparency can prove to be an obstacle for clinical adoption, where interpretability plays a vital role in establishing confidence among physicians. Despite the proposed ResNet-LSTM architecture showing remarkable performance on brain tumor progression prediction, it is imperative to subject it to critical introspection in terms of its limitations and challenges in practical implementation. One of the major limitations of the proposed architecture involves the possibility of it being

dataset-biased, having been trained on a relatively limited dataset. This not only poses the current challenge of lack of generalizability but also raises the prospect of the model not being universally applicable. Furthermore, in terms of interpretability, the proposed architecture, as a typical deep learning model, lacks clarity in its predictive capacities and can only be described as a black box. While its performance in terms of predictive accuracies appears promising, it lacks the major imperative of transparency in its predictive capacities. Furthermore, in terms of its practical implementation in a typical clinical setup, its implementation in a healthcare organization's computerized framework not only poses a number of challenges in terms of processing power but also faces hurdles in terms of approval. In comparison to the emerging trend involving the utilization of Vision Transformers and hybrid attention models in computer vision tasks, the proposed architecture outperforms in terms of spatial-temporal learning but requires

Finally, the system's performance remains heavily dependent on data quality and quantity. High-quality, well-labeled MRI datasets are necessary for reliable training and testing. In cases where data are scarce, inconsistent or poorly standardized, especially for rare tumor types, the model's accuracy and robustness may be compromised.

10. FUTURE ENHANCEMENTS

The suggested brain tumor development analysis system, through the integration of ResNet and LSTM networks, is a great step forward in medical imaging and diagnostic accuracy. With the integration of spatial feature extraction and temporal sequence modeling, the system proves to have a robust capacity for enhancing tumor classification and development detection in MRI and PET/CT scans. The application of deep learning techniques guarantees flexibility to intricate and heterogeneous medical information, rendering the system a solid and helpful resource for radiologists and physicians. In the future, a number of upgrades can further solidify the system's clinical significance and diagnostic capability. One of the promising avenues is incorporating more detailed clinical data, including patient history, genetic markers, and tumor metadata. This would allow for the creation of personalized prediction models, able to make predictions tailored to specific patients based on individual data instead of using imaging data. The inclusion of multi-modal datasets that consist of MRI, PET, CT, and molecular-level information might also enable a more comprehensive insight into tumor biology and progression.

Another possible upgrade is the use of explainable AI (XAI) methods to enhance model interpretability. This would enable clinicians to comprehend predictions more, leading to more trust and usability in actual medical practice. Additionally, broadening

the dataset to encompass a more diverse collection of tumor types, stages, and patient populations would address limitations in terms of generalizability and robustness.

The system could be evolved into real-time diagnostic tools by maximizing computational efficiency and embedding the model within hospital information systems. Such implementation would enable it to aid radiologists in routine clinical tasks, facilitating early detection and timely treatment planning. Further, the fusion of multi-modal imaging modalities like PET and CT scans can enhance diagnostic accuracy by the juxtaposition of functional and structural modalities. Future research would include further exploration into more sophisticated deep learning architectures such as Attention Mechanisms and 3D Convolutional Neural Networks (CNNs) for the processing of volumetric MRI scans to deliver even more accurate tumor detection and segmentation (Zhang et al., 2021). Such models can enhance the management of multi-dimensional brain imaging data, resulting in greater accuracy when it comes to predicting tumor progression. The addition of real-time diagnostic functionalities to the system, like hospital information system integration and radiology device integration, would allow for on-the-fly analysis and enable real-time decision-making. Also, the implementation of Explainable AI (XAI) methods would make medical experts understand the model's decisions better, raising confidence and clarity in clinical use. Although the proposed ResNet-LSTM architecture shows excellent performance characteristics in analyzing the progress of brain tumors, several modifications can be incorporated in the future to further increase its relevance and generalizability to actual applications. An initial and fundamentally more comprehensive step would be to test the architecture using larger, multi-institutional collections of images, such as BraTS, to further reduce the potential of overfitting. Next, methods of explainable AI (XAI) should be incorporated to further increase interpretability. A tool would be Grad-CAM's visualizations of heat maps. Other modifications to be incorporated in the future may include additional inputs related to the patients' and tumors' characteristics. A combination of all these modifications would collectively further increase the scalability and interpretability of the proposed architecture. From an overall systems integration standpoint, modifications to ensure higher computational efficiency would be necessary to facilitate its use in actual healthcare settings (Baid et al., 2021).

Lastly, the extension of the system to predictive analytics of tumor development and tracking of the response to therapy can assist clinicians in monitoring the efficacy of various therapeutic interventions. These extensions would render the system more versatile, dependable, and effective in diagnostic as well as therapeutic applications, further propelling the use of AI in medical imaging.

REFERENCES

- Akkus, Z., Galimzianova, A., Hoogi, A., Rubin, D. L., & Erickson, B. J. (2017). Deep learning for brain MRI segmentation: State of the art and future directions. *Journal of Digital Imaging*, 30(4), 449–459. DOI: 10.1007/s10278-017-9983-4 PMID: 28577131
- Ali, S., Li, J., Pei, Y., Khurram, R., & Rehman, K. ur, & Mahmood, T. (2022). A Comprehensive Survey on Brain Tumor Diagnosis Using Deep Learning and Emerging Hybrid Techniques with Multi-modal MR Image. *Archives of Computational Methods in Engineering*, 29(7), 4871–4896. DOI: 10.1007/s11831-022-09758-z
- Arabahmadi, M., Farahbakhsh, R., & Rezazadeh, J. (2022). Deep learning for smart Healthcare—A survey on brain tumor detection from medical imaging. *Sensors (Basel)*, 22(5), 1960. PMID: 35271115
- Baid, U., Ghodasara, S., Mohan, S., Bilello, M., Calabrese, E., Colak, E., Farahani, K., Kalpathy-Cramer, J., Kitamura, F. C., Pati, S., Prevedello, L. M., Rudie, J. D., Sako, C., Shinohara, R. T., Bergquist, T., Chai, R., Eddy, J., Elliott, J., Reade, W., & Schaffter, T. (2021). The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification. ArXiv.org. <https://arxiv.org/abs/2107.02314>
- Cheng, J., Huang, W., Cao, S., Yang, R., Yang, W., Yun, Z., Wang, Z., & Feng, Q. (2015). Enhanced performance of brain tumor classification via tumor region augmentation and partition. *PLoS One*, 10(10), e0140381. DOI: 10.1371/journal.pone.0140381 PMID: 26447861
- Isin, A., Direkoglu, C., & Sah, M. (2016). Review of MRI-based brain tumor image segmentation using deep learning methods. *Procedia Computer Science*, 102, 317–324. DOI: 10.1016/j.procs.2016.09.407
- Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A. W. M., van Ginneken, B., & Sánchez, C. I. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42, 60–88. DOI: 10.1016/j.media.2017.07.005 PMID: 28778026
- Mohsen, H., El-Dahshan, E. S. A., El-Horbaty, E. S. M., & Salem, A. B. M. (2018). Classification using deep learning neural networks for brain tumors. *Journal of Biomedical Informatics*, 81, 111–122. DOI: 10.1016/j.jbi.2018.04.001
- Nickparvar, M. (2021). Brain Tumor MRI Dataset. Kaggle.com. <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

- O'Shea, K., & Nash, R. (2015). An introduction to convolutional neural networks. arXiv. <https://arxiv.org/abs/1511.08458>
- Prabukumar, M., Agilandeewari, L., & Ganesan, K. (2019). An intelligent brain tumor diagnosis system using cuckoo search optimization and support vector machine classifier. *Journal of Ambient Intelligence and Humanized Computing*, 10, 373–388. DOI: 10.1007/s12652-018-0786-5
- Rehman, A., Naz, S., Razzak, M. I., Akram, F., & Imran, M. (2019). A deep learning-based framework for automatic brain tumors classification using MRI scans. *Machine Vision and Applications*, 30, 305–318. DOI: 10.1007/s00138-018-00929-2
- Saleh, Abbadullah. H., Atila, Ü., & Menemencioğlu, O. (2024). Multimodal Fusion for Enhanced Semantic Segmentation in Brain Tumor Imaging: Integrating Deep Learning and Guided Filtering Via Advanced 3D Semantic Segmentation Architectures. *International Journal of Imaging Systems and Technology*, 34(5). Advance online publication. DOI: 10.1002/ima.23152
- Sartaj. (2024). Brain Tumor Classification (MRI). Kaggle.com. <https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri>
- Shen, D., Wu, G., & Suk, H.-I. (2017). Deep learning in medical image analysis. *Annual Review of Biomedical Engineering*, 19, 221–248. DOI: 10.1146/annurev-bioeng-071516-044442 PMID: 28301734
- Sinha, A., & Verma, R. K. (2019). Automated detection of brain tumors using deep learning techniques: A survey. *Procedia Computer Science*, 167, 2039–2048. DOI: 10.1016/j.procs.2020.03.251
- Vatanpour, M., Haddadnia, J., & Bajestani, S. S. (2025). Combination of Graph and Convolutional Networks for Brain Tumor Segmentation from Multi-Modal MR Images In Clinical Applications. *Frontiers in Biomedical Technologies*.
- Zhang, W., Wu, Y., Yang, B., Hu, S., Wu, L., & Dhelim, S. (2021). Overview of Multi-Modal Brain Tumor MR Image Segmentation. *Health Care*, 9(8), 1051. DOI: 10.3390/healthcare9081051 PMID: 34442188
- Zhao, W., Jiang, Y., Yang, H., & Zhao, Y. (2021). Brain tumor segmentation using hybrid 3D–2D deep learning model with limited dataset. *Medical Image Analysis*, 66, 101851. DOI: 10.1016/j.media.2020.101851
- Zhou, Z., Rahman Siddiquee, M. M., Tajbakhsh, N., & Liang, J. (2020). UNet++: Redesigning skip connections to exploit multiscale features in image segmentation. *Frontiers in Neuroscience*, 14, 699. DOI: 10.3389/fnins.2020.00699 PMID: 31841402